

Andrea Zingarofalo

Curriculum Vitae et Studiorum

About Me

Theoretical physicist exploring how fundamental physical laws shape the limits of knowledge, communication, and computation. My current research focuses on device-independent methods for certifying quantum resources, a crucial task for establishing trust and security in next-generation quantum technologies.

Education

2025 – 2029 **PhD in Quantum Information Theory** *en cotutelle internationale*, *Université libre de Bruxelles & Université Paris-Saclay*
- Hosted by the Laboratoire d'Information Quantique (ULB) and the Institut de Physique Théorique (IPhT, CEA Paris-Saclay)
- Thesis: *Self-testing and device-independent verification of quantum computation*
- Advisors: Prof. S. Pironio (ULB) & Prof. J.-D. Bancal (IPhT)

2020 – 2025 **Diploma di Licenza**, *Advanced School ISUFI* (Excellence Program), 70/70 cum laude
- ∼10% Acceptance rate for the Natural Sciences Class in 2020
- Modules: Group Theory and Symmetries, Celestial Mechanics, Astroparticle Physics, Applied Physics I–II, Philosophy of Science, ICT for Research, Public Speaking, Scientific Writing

2023 – 2025 **MSc in Theoretical Physics**, *University of Salento*, 110/110 cum laude, GPA: 29.8/30
- Modules: Quantum Field Theory I–II, QCD, Statistical Mechanics, Quantum Computing, Physics of Complex Systems, Data Analysis, Mathematical Methods for AI
- Thesis: *Exploring the boundaries of quantum correlations: Bell inequalities, convex duality, and self-testing*
- Advisors: Prof. L. Martina & Prof. J.-D. Bancal

2024 – 2025 **Exchange program**, *École normale supérieure*, Master ICFP (International Center for Fundamental Physics) M2, Theoretical Physics track
- Modules: General Relativity, Quantum Field Theory, Statistical Field Theory, Algorithms & Computations

2020 – 2023 **BSc in Physics**, *University of Salento*, 110/110 cum laude, GPA: 29.9/30
- Electives: Nuclear Physics, Astronomy, Foundations of Physics
- Thesis: *Study of one-dimensional quantum integrable systems via the Bethe Ansatz*
- Advisor: Prof. L. Martina

2015 – 2020 **High School Diploma**, *Liceo Scientifico V. Lilla*, Oria, Italy, 100/100 cum laude
- Activities: Student ambassador

Experience

2026 – 2029 **F.R.S. - FNRS Research Fellow**, *Laboratoire d'Information Quantique – Université libre de Bruxelles*
- Awarded the highly competitive F.R.S.-FNRS *Aspirant* Fellowship (acceptance rate <20%), providing full funding for my entire PhD
- Research on self-testing and device-independent protocols for certifying components of quantum computation, from multipartite entanglement to error-correcting codes
- Supervisor: Prof. S. Pironio

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2026 – 2027 **Teaching Assistant**, *Université libre de Bruxelles*
- 24 h of tutorials for the Master's course PHYS-F509 – *Quantum Information Theory* (Prof. S. Pironio)

Oct 2025 – Feb 2026 **Visiting PhD Student**, *Institut de Physique Théorique – CEA Paris-Saclay*
- Research on verification protocols for quantum computation
- Supervisor: Prof. J.-D. Bancal

Nov 2025 – ongoing **AI Model Training Specialist (Physics)**, *Mindrift*
- Contributed to the training and evaluation of LLM models by providing expert-level physics supervision and feedback
- Analyzed model outputs to identify systematic failure modes and guide iterative model refinement

Mar – Jul 2025 **Research Intern**, *Institut de Physique Théorique – CEA Paris-Saclay*
- Derived analytically new Bell inequalities for partially entangled states
- Implemented SDP relaxations to numerically estimate quantum bounds of Bell inequalities via the Navascués-Pironio-Acín hierarchy
- Supervisor: Prof. J.-D. Bancal

Mar – Jul 2024 **Master's Student Researcher**, *Mathematical Physics Group – University of Salento*
- Derived multi-kink solutions of a generalized Fokker-Planck equation via Hirota's bilinear method
- Developed automated Mathematica scripts for symbolic generation of soliton solutions
- Supervisor: Prof. L. Martina

2026 – Present **Science Communicator**, *Dear Scientist*
- Volunteer pen-pal scientist for Italian elementary and middle school students, guiding them through the scientific method with simple experiments rather than direct answers
- Activity carried out under the patronage of the Consulate General of Italy in Boston

2019 – Present **Private Tutor** in Mathematics, Physics, and Chemistry for high-school and university students, including A-Levels and Oxford/Cambridge admission interviews

Publications

Journal Articles

[1] F. Giglio, G. Landolfi, L. Martina, and **A. Zingarofalo**, "Integrability properties and multi-kink solutions of a generalised Fokker-Planck equation", *J. Phys. A: Math. Theor.* **58**, 165202 (2025).

Preprints and Work in Preparation

[1] M.-E. Liu, **A. Zingarofalo**, V. Barizien, and J.-D. Bancal, "All Bell inequalities with a quantum advantage in the minimal scenario are self-testing", arXiv:2611.06997 (2026).

[2] **A. Zingarofalo**, J.-D. Bancal, and S. Pironio, "Towards photonic Bell tests with minimal sample complexity", *in preparation* (2026).

Outreach

[1] **A. Zingarofalo**, "Quantum correlations: A geometric perspective", in *Rete di Idee – XIII Edizione, Atti del Convegno, Catania 2025*, M. F. Anfuso and B. D'Arrigo, eds., Scuola Superiore di Catania, pp. 91–131 (2026).

[2] **A. Zingarofalo**, "Il conflitto tra meccanica quantistica e senso comune", *Il Chiasmo*, Treccani – The Italian Encyclopaedia Institute (2024).

[3] **A. Zingarofalo**, "Il ruolo delle simmetrie nel problema di Keplero", in *Rete di Idee, Udine 2022*, D. Avaro and F. Decataldo, eds., Forum Editrice, Udine, pp. 81–100 (2023).

Schools and Conferences

Conferences

2026
- Young Quantum Information Science 2026, TU Wien, Vienna, 24–28 Aug
  - Poster: Towards photonic Bell tests with minimal sample complexity
- 851. WE-Heraeus-Seminar: 90 Years of EPR Paradox, Deutsche Physikalische Gesellschaft, Physikzentrum Bad Honnef, 7–10 Apr
  - Poster: Towards photonic Bell tests with minimal sample complexity
- VERIqTAS Workshop 2026, ENS Lyon, 23–25 Mar

2025 ○ Rete di Idee XIII, Scuola Superiore di Catania, 21–23 Nov
  - Invited Talk: From Bell's theorem to quantum technologies via the device-independent paradigm
○ Workshop in honor of I. Cirac and P. Zoller, Académie royale de Belgique, Brussels, 3 Oct
○ Interdisciplinary Statistical Physics workshop, University of Salento, Lecce, 18–19 Sep

2022 ○ Rete di Idee X, Superiore Universitaria di Udine, 7–9 Oct
  - Invited Talk: The role of symmetry in Kepler's problem

Domain-Specific Schools

2026 ○ Quantum Key Distribution Summer School 2026, SwissMAP & ETH Zürich, Les Diablerets, 16–21 Aug
  - Poster: Towards photonic Bell tests with minimal sample complexity
○ Quantum Technologies for Computation and Communication Summer School, Institut d'Etudes Scientifiques de Cargèse, 8–20 Jun
  - Poster: Towards photonic Bell tests with minimal sample complexity

2025 ○ Qiskit Global Summer School, IBM, 7–22 Jul
○ Mathematical Aspects of Quantum Information Summer School, Université Paris-Saclay & Institut Polytechnique de Paris, Paris, 16–20 Jun
  - Poster: Orthogonal faces in the CHSH scenario

2024 ○ Theoretical Physics of Fundamental Interactions Summer School, Scuola Normale Superiore, Pisa, 27–31 May

Others

2025 Summer School "Reframing Sustainability: Bridging Environmental, Energy and Economical Challenges", Advanced School ISUFI, Otranto, 10–12 Sep

2023 Winter School "Green and Digital Transition in Europe", ENCATC (European Network of Cultural Administration Training Centres), Brussels, 2–7 Dec

2019 Physics Summer School, University of Salento, Lecce, 2–6 Sep

2019 ○ University Orientation School, Scuola Normale Superiore and Accademia Nazionale dei Lincei, Rome, 8–13 Jul

2018 ○ Stage at Asiago Astrophysical Observatory, INAF (Italian Institute of Astrophysics) and University of Padua, Asiago, 9–13 Jul

2016 ○ Summer School on Astronomy, INAF and SAIt (Italian Astronomical Society), Sant'Agata sui Due Golfi, 5–8 Jul

Awards and Scholarships

2026 ○ Selected as Mentee of the *LeadTheFuture* program, a free mentorship program for STEM talents (acceptance rate below 15%)
○ Awarded the ADI cotutelle doctoral fund of Université Paris-Saclay (declined in favour of the F.R.S.-FNRS *Aspirant* Fellowship)
○ Winner of the *Soliani Prize* of the University of Salento for the best Master's thesis of 2025 in Theoretical Physics, Mathematical Physics and Biomathematics (€ 1500 gross)
○ Winner of the *Nova 111 Student List* (Italy), Mathematics, Data Analytics & Physics category – selected among 111/1870 applicants nationwide by Nova Talent, an international network of top emerging talent in STEM and business

2025 ○ Winner of RIASISSU Rete di Idee, XIII edition, Catania
○ Selected among the Top 100 STEM graduates in Italy (Class of 2025), by Almalaurea, the national consortium of Italian universities
○ Winner of an ISUFI Research & Innovation grant for international research mobility (€ 5000)

2024 ○ Winner of a Cité Internationale Universitaire de Paris scholarship (accommodation)

2022 ○ Winner of RIASISSU Rete di Idee, X edition, Udine

2020 ○ Winner of the ISUFI Scholarship at the University of Salento, awarded for academic excellence (full accommodation and a stipend)

2018 ○ Senior finalist at XVI Italian Astronomy Olympiads, ranking among the top 30/6000 participants

Technical Skills

Programming and Scientific Computing

Programming Python (`NumPy`, `SciPy`, `Matplotlib`, `pandas`), `C`/`C++`, Julia, Bash; MATHEMATICA

Quantum Libraries Quantum simulation with `Qiskit` and `PennyLane`; SDPs and convex optimization using `CVXPY`, `MOSEK`, `ncpol2sdpa`

Data Analysis ROOT framework for experimental data, and machine learning basics with `scikit-learn` and `TensorFlow`

Workflow LaTeX, Markdown, `git` version control, Inkscape, Microsoft Office; comfortable with UNIX systems, shell scripting, and use of cluster (CECI cluster, ULB)

Languages

Italian (native), English (C1), French (B2), German (A2)

Memberships

European Physical Society (2026-), IEEE Quantum Technical Community (2025-), Italian Physical Society – SIF (2023-), RIASISSU Excellence Network (2023-2025)

Additional Information

Music Guitar Grade 2 (Distinction), Trinity College of London

Interests Outside academia, I'm a tech enthusiast with a passion for music, history, travelling and geopolitics. I am also an amateur astronomer.